Name:_____

Exam Question Booklets
OCR Physics A

Booklet 13
15 Questions
122 marks

Things to remember:

✓

✓

✓

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✓

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✓

1

An aerial mounted vertically transmits vertically polarised radio waves of frequency 1.0×10^9 Hz. The waves are detected by a receiving aerial some distance away. Initially the receiving aerial is also mounted vertically as shown in Fig. 5.2.

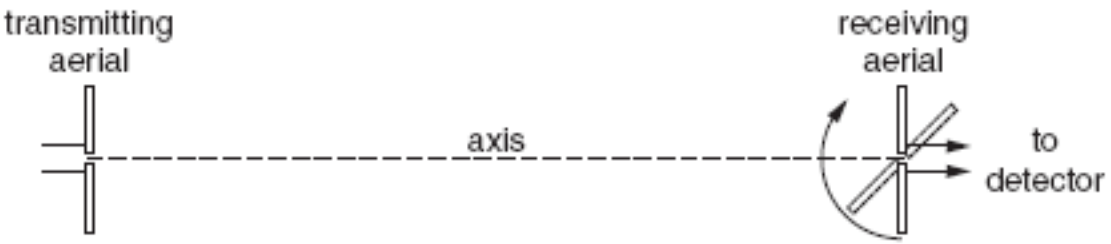

Fig. 5.2

The length of each aerial is half the wavelength of the radio waves.

(i) Calculate the wavelength of the waves.

wavelength = ... m **[2]**

(ii) Calculate the length of an aerial.

length = ... m **[1]**

(iii) The receiving aerial is rotated through 180° about the axis joining the centres of the two aerials. See Fig. 5.2. Describe and explain how the output signal from the receiving aerial changes with the angle of rotation.

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.....

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.....

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..... **[3]**

2

(a) Fig. 6.1 shows an electrical circuit.

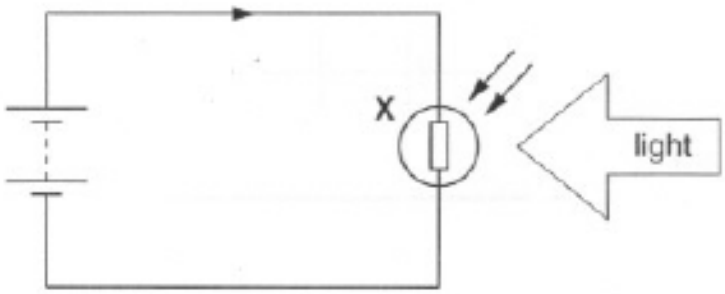

Fig. 6.1

The battery has negligible internal resistance.

(i) Name the component **X**.

.....[1]

(ii) Explain what happens to the current in the circuit when the intensity of the light incident on the component **X** is increased.

.....

.....

.....[2]

(iii) Describe what happens to the current in the circuit when the intensity of the light is kept the same, but a battery of half the e.m.f. is used.

.....

.....[1]

(a) **(i)** State the unit of electric charge.

..... [1]

(ii) Name an instrument that may be used to measure the potential difference (p.d.) across an electrical component.

..... [1]

(b) A 36 W lamp draws a constant current of 3.0 A over a period of 600 s from a battery. Calculate

(i) the p.d. across the lamp,

p.d. = ... V [3]

(ii) the energy transferred by the lamp,

energy = ... J [2]

(iii) the charge passing through the lamp,

charge = ... C [3]

(iv) the number of electrons passing through the lamp.

number = ... [2]

4

(a) Draw a labelled diagram to show the arrangement that you would use to produce, on a screen, a double-slit interference pattern for light.

[1]

In order to see the interference pattern clearly in a darkened room, suggest suitable values for

(i) the distance from the double-slit to the screen

distance = ...

(ii) the distance between the centres of the slits in the double-slit.

distance= .....

[2]

(b) Fig. 4.1 shows the light and dark fringes in a typical double-slit interference pattern. The diagram is drawn to full scale.

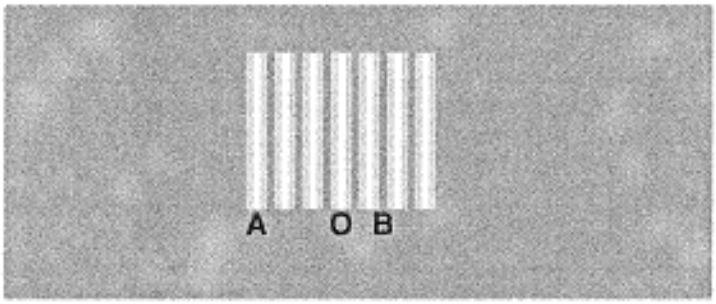

Fig. 4.1

(i) By taking measurements directly from Fig. 4.1, determine the fringe separation x.

x = ...mm [1]

(ii) Fringe **O** is the bright fringe at the centre of the pattern, **A** is a bright fringe and **B** is a dark fringe. For each of these fringes, state in terms of the wavelength λ, the value of the path difference for light coming from each slit.

fringe **O** path difference = ...

fringe **A** path difference = ...

fringe **B** path difference = ...

[3]

(c) Use the formula for double-slit interference to explain how the fringe separation would change if blue light were used instead of red light.

.....

.....

.....[2]

Fig. 3.2 is a plan view of the same arrangement shown in Fig. 3.1 with the addition of a metal plate **M** placed in front of the transmitter. The plate **M** contains a double slit.

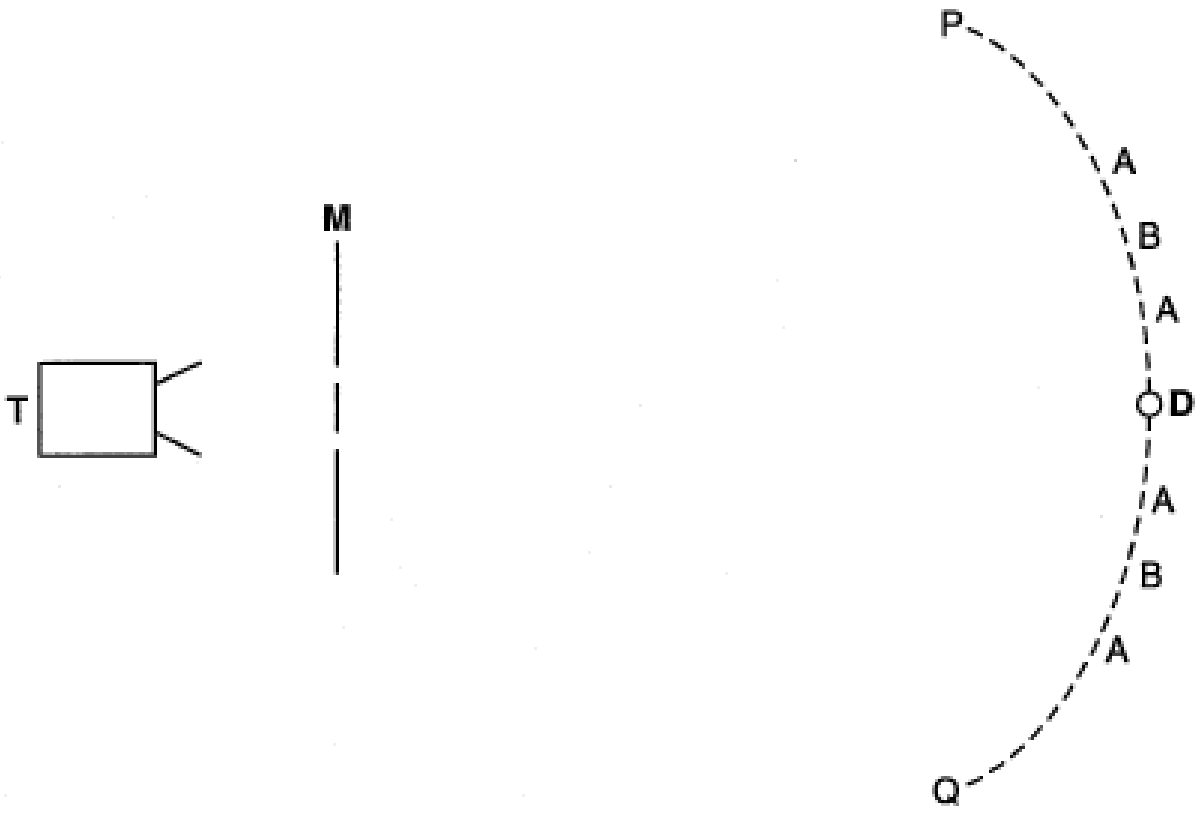

Fig. 3.2

When the detector **D** is placed in the position shown, the meter reading is a maximum, but as it moves along the horizontal arc PQ the reading passes through a sequence of low and high readings at positions A and B respectively.

(i) State the name of the phenomenon that accounts for this.

.....[1]

(ii) Explain why the meter reading is a maximum when the detector is in the position shown (i.e. directly opposite the centre of the double slit).

.....

.....

.....[2]

(iii) Explain

1. why the meter reading is low at positions A

.....

.....

2. why the meter reading is high at positions B.

.....

.....[3]

(iv) Suggest, with reasons, how the total number of low and high readings along the arc PQ changes when the separation between the slits in M is **reduced**.

.....

.....

.....[2]

Fig. 4.1 shows an electrical circuit for a small dryer used to blow warm air.

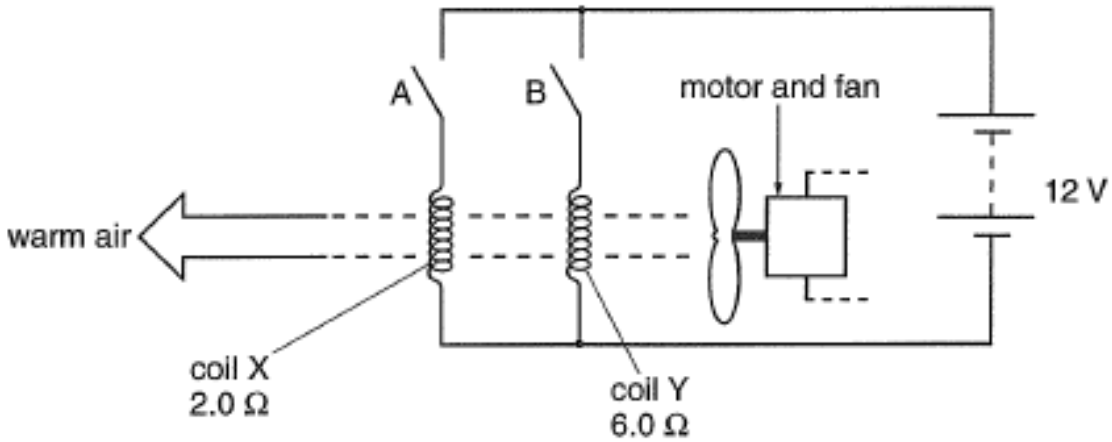

Fig. 4.1

It may be assumed that coils X and Y have constant resistances of 2.0 Ω and 6.0 Ω respectively. The 12 V supply has negligible internal resistance.

(a) Calculate the total resistance of the circuit with both switches closed.

resistance = ... Ω [3]

(b) Calculate the power dissipated by the coil X with only switch A closed.

power = ... unit = ... [4]

Fig. 3.1 shows a cell of e.m.f. E and internal resistance r connected to a variable resistor.

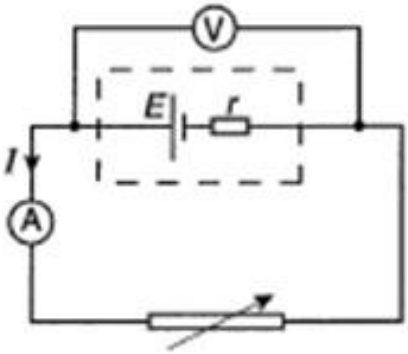

Fig. 3.1

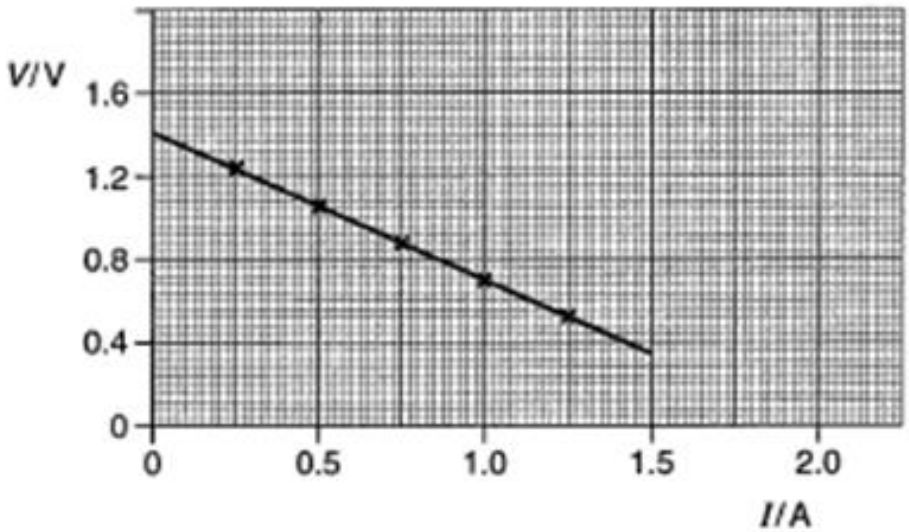

Fig. 3.2

Fig. 3.2 shows the variation of the p.d. V across the terminals of the cell with the current I drawn from the cell.

(a) Explain how Fig. 3.2 shows that the e.m.f. E is 1.4 V.

.....

..... [1]

(b) **(i)** Use Fig. 3.2 to determine the maximum possible current that can be drawn from the cell.

current = ... A [1]

(ii) Calculate the internal resistance r of the cell.

r = ... Ω [2]

(iii) Suggest why it may not be advisable to maintain the current determined in **(b)(i)** for a long time.

.....

..... [1]

(a) **(i)** State **three** phenomena that apply to all transverse and longitudinal waves.

1. .....

2. .....

3. ..... [2]

(ii) State a wave phenomenon that applies to transverse waves only.

..... [1]

(b) Fig. 5.1 shows an arrangement that can be used to determine the wavelength of microwaves.

Fig. 5.1

Microwaves leave the transmitter and move in a direction **TP** which is at right angles to the metal plate. A standing (stationary) wave is formed between **T** and **P**.

(i) State what is meant by a *standing wave* and explain how it is formed in this case.

.....

.....

.....

.....

.....

.....

..... [3]

(ii) When a small microwave detector **D** is moved slowly from **T** towards **P** the signal received changes from strong to weak to strong to weak etc. The distance between the positions of neighbouring weak signals is 1.4 cm.

Calculate for these microwaves

1 the wavelength

wavelength = ... cm [1]

2 the frequency.

frequency = ... Hz [2]

(iii) Describe how you could test whether the microwaves leaving the transmitter were plane polarised.

.....

.....

.....

.....

..... [2]

(a) With reference to the vibrations involved, state the difference between *transverse* and *longitudinal* waves. State an example of each.

.....

.....

.....

example of a transverse wave: .....

example of a longitudinal wave: .....[3]

(b) State one similarity and one difference between *progressive* waves and *standing (stationary)* waves.

similarity: .....

.....

difference: .....

.....[2]

(c) Explain what is meant by

(i) a *node*

.....

.....[1]

(ii) an *antinode*.

.....

.....[1]

(d) The distance between a node and the neighbouring antinode in a standing wave formed on a stretched string is 0.12 m. Calculate the wavelength.

wavelength = ... m [1]

(e) Describe and explain how a standing wave can be formed using a suitable source of longitudinal waves. State the wave source and describe the arrangement using a labelled diagram. Label the position of a node (label as **N**) and an antinode (label as **A**).

.....

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.....

.....

.....[4]

Fig. 5.1 shows a potential divider circuit used to monitor the temperature of a greenhouse.

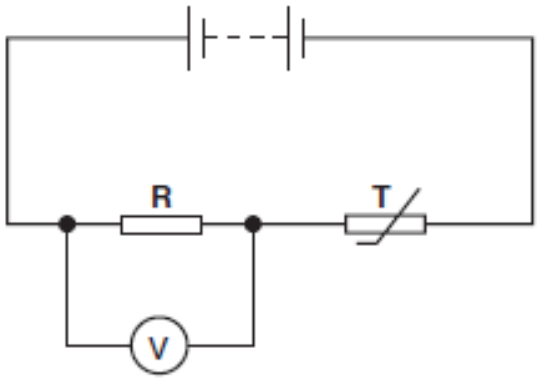

Fig. 5.1

The thermistor **T** is a negative temperature coefficient type. The voltmeter is placed across the resistor **R**. Describe and explain how the voltmeter reading changes as the temperature of the greenhouse **increases**.

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.....[4]

Fig. 5.1 shows a potential divider circuit designed as a touch-sensor.

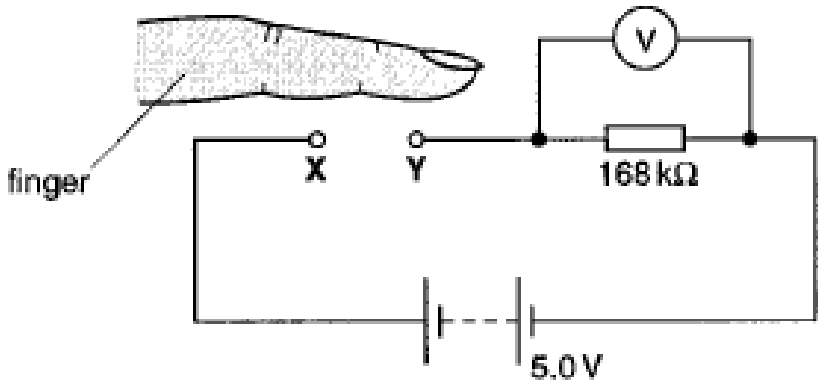

Fig. 5.1

The battery has negligible internal resistance and the voltmeter has infinite resistance.

(a) Explain why the voltmeter reading is zero when there is nothing connected between the contacts **X** and **Y**.

.....

.....[1]

(b) When the finger makes contact between **X** and **Y**, the voltmeter reading changes from 0 V to 3.4 V because of the electrical resistance of the skin. Use this information to calculate the electrical resistance of the skin between the two contacts.

resistance = ... kΩ [3]

(a) Kirchhoff's first law is based on the conservation of an electrical quantity. State the law and the quantity conserved.

.....

.....

..... [2]

(b) Fig. 4.1 shows a potential divider circuit. The battery has negligible internal resistance and the voltmeter has very high resistance.

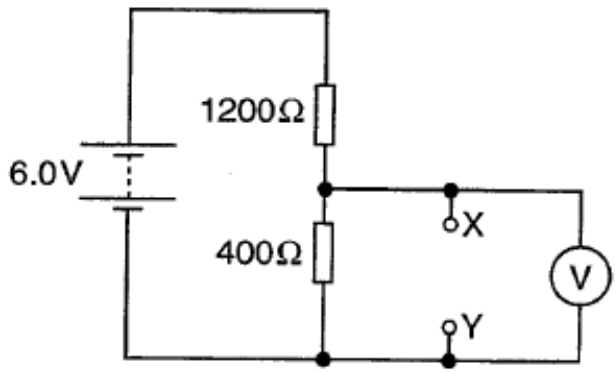

Fig. 4.1

(i) Show that the voltmeter reading is 1.5 V.

[2]

(ii) An electric device rated at 1.5 V, 0.1 A is connected between the terminals X and Y. The device has constant resistance. The voltmeter reading drops to a very low value and the device fails to operate, even though the device itself is not faulty.

1. Calculate the total resistance of the device and the 400 Ω resistor in parallel.

resistance = ... Ω [3]

2. Calculate the p.d across the device when it is connected between X and Y.

p.d. = ... V [2]

3. Why does the device fail to operate?

..... [1]

(a) State **two** features common to all types of radiation in the electromagnetic spectrum.

.....

.....

.....

..... [2]

(b) A radioactive material emits photons, each having an energy of 1.0 MeV.

(i) Explain what is meant by a *photon*.

.....

..... [1]

(ii) Show that the energy of each photon is 1.6×10^{-13} J.

[2]

(iii) Calculate the frequency of the electromagnetic radiation emitted by the radioactive material.

frequency = ... unit = ... [4]

(iv) Calculate the wavelength λ of the radiation.

λ = ... m [2]

(v) State the principal type of electromagnetic radiation emitted by the material.

..... [1]

(a) Electrons are emitted from the surface of zinc when it is exposed to ultraviolet radiation.

(i) Name this phenomenon.

.....[1]

(ii) State a typical value for the wavelength of ultraviolet radiation in metres.

.....[1]

(b) Electromagnetic radiation incident on a metal plate releases energetic electrons from its surface. The metal plate is placed in an evacuated chamber. The energy of each photon is 2.8 eV. The metal has a work function energy of 1.1 eV.

(i) Explain what is meant by the *work function energy* of the metal.

.....

.....[1]

(ii) State the speed of the photons.

.....[1]

(iii) For an electron emitted from the surface of the metal, calculate

1. its maximum kinetic energy in joules

energy = ... J [3]

2. its maximum speed.

speed = ... $m\,s^{-1}$ [2]

(iv) State the change, if any, to your answer for the maximum speed of an electron emitted from the surface of the metal when the intensity of the incident electromagnetic radiation is doubled.

.....

.....[1]

15 A negatively charged metal plate is exposed to electromagnetic radiation of frequency f. Fig. 7.1 shows the variation with f of the maximum kinetic energy E_k of the photoelectrons emitted from the surface of the metal.

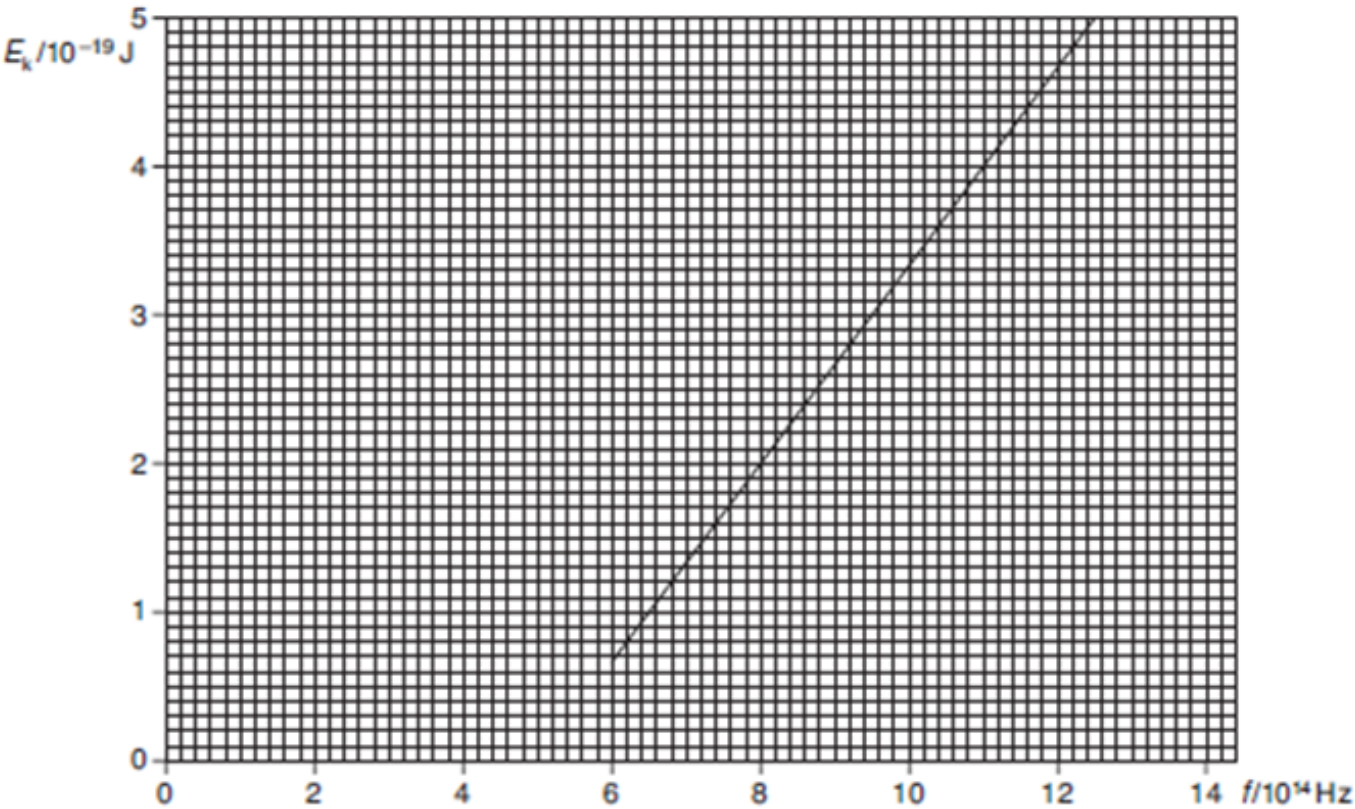

Fig. 7.1

(i) Define the *threshold frequency* of a metal.

.....

.....[1]

(ii) **1** Explain how the graph shows that the threshold frequency of this metal is 5.0×10^{14} Hz.

.....

.....[1]

2 Calculate the work function energy of this metal in joules.

work function energy = ... J [2]

(iii) Electromagnetic radiation falls on the surface of a metal having work function energy greater than your answer in **b(ii)**.

1 State and explain the change, if any, to the gradient of the line shown in Fig. 7.1.

.....

.....

.....[2]

2 State and explain the change, if any, to the position of the line shown in Fig. 7.1.

.....

.....

.....[2]